SAR Automatic Target Recognition Based on Dictionary Learning and Joint Dynamic Sparse Representation

Yongguang Sun, Lan Du, *Member, IEEE*, Yan Wang, Yinghua Wang, and Jing Hu

***Abstract*—In this letter, we propose a novel automatic target recognition (ATR) method based on dictionary learning and joint dynamic sparse representation (DL-JDSR) for synthetic aperture radar (SAR) images. First, in the feature extraction step, we extract two kinds of features, i.e., the image domain amplitude feature and the scale-invariant feature transform (SIFT) feature, of which the image domain amplitude feature describes intensity information and the SIFT feature describes gradient information. These two features will be jointly utilized to combine the two kinds of information for SAR ATR. Second, we introduce the dictionary-learning method, the label-consistent K-singular value decomposition, into the training step to learn dictionaries for the two features rather than directly using all training samples as the fixed dictionaries in the traditional sparse representation method. The learned dictionaries have smaller sizes and are more distinctive among different classes, which can speed up our recognition and improve the accuracies. Third, the JDSR algorithm used in the test step employs a more flexible atom selection method, which enables the two features from an image data to share the similar but not exactly the same sparse mode. Experiments on the moving and stationary target acquisition and recognition data set show that the proposed method is an effective way to recognize SAR images.**

***Index Terms*—Automatic target recognition (ATR), joint dynamic sparse representation (JDSR), label-consistent K-singular value decomposition (K-SVD) (LC-KSVD), synthetic aperture radar (SAR).**

I. Introduction

WITH the development of the synthetic aperture radar (SAR) imagery technology, automatic target recognition (ATR) for SAR images has attracted more and more attention. Lots of methods have been applied, such as principal component analysis (PCA) [1], template matching [2], etc. PCA projects samples into another subspace to gain distinctive features, and the test samples are recognized with a well-trained kernel support vector machine (SVM). In template matching, a test sample is matched with training samples, and the criterion of the minimum square error is used as the recognition rule. They have achieved good recognition results for SAR images.

Sparse representation has aroused an increasing interest in recent years. In [3], a method based on sparse representation for face recognition is proposed, and robust results are achieved even under the circumstances of occlusion and noise. Later, sparse representation is introduced into SAR ATR. Zhang *et al.* [4] proposed a novel joint sparse representation (JSR) method for SAR ATR, which takes advantage of correlations among multiple views of the same target from different aspects and achieves impressive recognition results.

Although good results have been obtained, there exist some problems when these methods are applied. First, the ATR performance is limited since only the amplitude information is used in traditional methods. Second, dictionaries used in sparse representation are usually made up of all training samples, leading to a great cost in storage and computation. Third, in the JSR method in [4], the restriction is so rigid that nonzero sparse coefficients of multiple views should be in the exactly same position. In addition, the difficulty to obtain multiple views of the same target limits the method's application in reality.

The scale-invariant feature transform (SIFT) descriptors, which are invariant to image scaling and rotation, have shown great power in the fields of object matching [5] and image registration [6]. The SIFT feature, generated with local SIFT descriptors, describes the gradient information of an image data, which can be a complement to the intensity information.

The K-singular value decomposition (K-SVD) [7] algorithm is proposed to learn compact dictionaries. However, as an unsupervised dictionary leaning method, K-SVD cannot utilize the label information. Jiang *et al.* [8] proposed label-consistent K-SVD (LC-KSVD) algorithm to learn discriminative dictionary for recognition. More specifically, the algorithm associates label information with each dictionary atom to enforce discriminability during the dictionary-learning process.

A joint dynamic sparse representation (JDSR)-based method for visual recognition is proposed in [9]. The method emphasizes that the multiple sparse representation vectors share joint sparse pattern at class level while allowing distinct sparse patterns at the atom level within each class in order to facilitate a flexible representation.

Inspired by [5]–[9], a method for SAR target recognition based on dictionary learning and joint dynamic sparse representation (DL-JDSR) is proposed in this letter. The main

Manuscript received March 31, 2016; revised July 8, 2016; accepted July 30, 2016. Date of publication September 30, 2016; date of current version December 7, 2016. This work was supported in part by the National Science Foundation of China under Grant 61271024 and Grant 61322103, by the Foundation for Doctoral Supervisor of China under Grant 20130203110013, and by the Science Foundation of Shaanxi Province under Grant 2015JZ016.
(Corresponding author: Lan Du.)
The authors are with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China, and also with the Collaborative Innovation Center of Information Sensing and Understanding, Xidian University, Xi'an 710071, China (e-mail: dulan@mail.xidian.edu.cn).
Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.
Digital Object Identifier 10.1109/LGRS.2016.2608578

contributions are summarized as follows: 1) amplitude feature and SIFT feature are jointly used in recognizing process; 2) compact and distinctive dictionaries are learned for JSR; and 3) a more flexible atom selection process is employed via the JDSR method. Experiment results on moving and stationary target acquisition and recognition (MSTAR) public data set show the effectiveness of the proposed method applied to SAR ATR.

The rest of this letter is organized as follows. In Section II, we introduce our method in three steps. In Section III, experiments are carried out to evaluate the performance of the proposed method and compare its performance with other methods. Finally, we conclude this letter in Section IV.

II. Methodology

A. Feature Extraction

Traditional SAR ATR methods [1], [2], [4] only utilize the amplitude feature, which limits the performance. Here, we will introduce a new feature that can be complementary to the amplitude feature and jointly use them in the recognizing process.

The amplitude feature is sensitive to the target position. Therefore, preprocessing before feature extraction is necessary, which includes segmentation, registration, and cutting. Segmentation [10] is to locate the target position and registration [11] is to put the target at the center of the image. Finally, the middle block with the size of 65 $\times$ 65 for each image data is used in the following experiments. We obtain the amplitude feature vector by directly reshaping all the pixel values into a long vector.

The SIFT descriptors are first proposed to perform reliable matching between different views of an object or scene [5]. Here, dense points instead of keypoints are used based on the fact that dense features work better for classification [12]. Specifically, dense feature points are obtained from the image using an overlapping, sliding window over the image. The extraction procedure of a local SIFT descriptor for a single feature point is shown in Fig. 1. First, the gradient magnitudes and orientations are computed at each image pixel in a region around the feature point. Second, these gradients are accumulated into orientation histograms each with 8 bins summarizing the contents over the subregions. Third, the histograms for the subregions are put together to form the local descriptor. Fig. 1 shows a 2 $\times$ 2 histogram array computed from an 8 $\times$ 8 set of pixels, whereas the experiments in this letter use a 4 $\times$ 4 histogram array computed from a 16 $\times$ 16 set of pixels. Therefore, the dimension of a local descriptor is $4\times4\times8=128$.

Since each local descriptor has a high dimension of 128, the SIFT feature of a SAR image will be an extremely long vector if we directly put all the local descriptors together. To overcome the difficulties, a three-level spatial pyramid with pooling [13] is used to generate the global feature with the typical pooling grids of 4 $\times$ 4, 2 $\times$ 2, and 1 $\times$ 1.

The extracted SIFT feature describes the gradient information, which can be an effective complementary to the intensity information that the amplitude feature describes. The extracted amplitude feature and the SIFT feature will be jointly used in the following recognizing process.

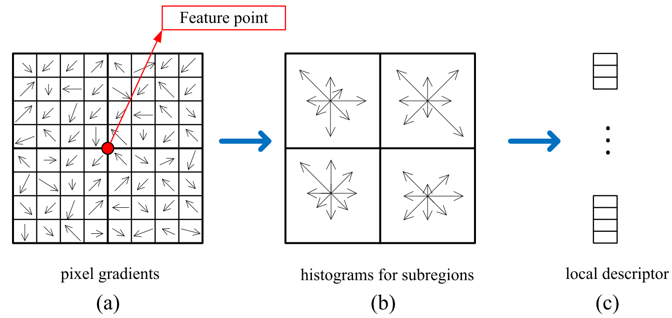

Fig. 1. Extraction procedure of a local SIFT descriptor for a feature point.

After feature extraction, the two features of all training samples are put together to form feature matrixes $\boldsymbol{F}^1 \in R^{d_1\times N}$ and $\boldsymbol{F}^2 \in R^{d_2\times N}$, where d_1 and d_2 denote the dimensions of the amplitude feature and the SIFT feature, and N is the number of training samples.

B. LC-KSVD Algorithm

Feature matrixes can be directly used as dictionaries for JDSR. However, they contain much redundancy and require large memorial space. More compact dictionaries can be acquired by dictionary-learning algorithms, of which the classic K-SVD algorithm [7] can be written as

$$\langle \boldsymbol{A}, \boldsymbol{X}\rangle = \arg\min_{\boldsymbol{A},\boldsymbol{X}} \|\boldsymbol{F}-\boldsymbol{A}\boldsymbol{X}\|_2^2 \quad \text{s.t. } \forall\, i, \|\boldsymbol{x}_i\|_0 \le S \tag{1}$$

where $\|\cdot\|_2$ represents l_2-*norm*, $\|\cdot\|_0$ represents l_0-*norm*, $\boldsymbol{F}=[\boldsymbol{f}_1, \boldsymbol{f}_2, \ldots, \boldsymbol{f}_N] \in R^{d\times N}$ is the input feature matrix and $\boldsymbol{f}_i$ is the feature vector of the ith sample, $\boldsymbol{A}=[\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_M] \in R^{d\times M}$ denotes the learned dictionary and $\boldsymbol{a}_m$ is the mth dictionary atom, $\boldsymbol{X}=[\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in R^{M\times N}$ denotes the sparse coefficient matrix and $\boldsymbol{x}_i$ is the sparse coefficient of the ith sample, and $\|\boldsymbol{F}-\boldsymbol{A}\boldsymbol{X}\|_2^2$ denotes the reconstruction error.

As an unsupervised dictionary-learning method, the K-SVD algorithm cannot exploit label information of training samples. In this letter, we introduce the LC-KSVD algorithm [8] to take advantage of the label information during dictionary-learning process. In this algorithm, a new label consistency constraint called "discriminative sparse-code error" is introduced and combined with the reconstruction error to form a unified objective function. It can be formulated as

$$\langle \boldsymbol{A}, \boldsymbol{X}\rangle = \arg\min_{\boldsymbol{A},\boldsymbol{X}} \|\boldsymbol{F}-\boldsymbol{A}\boldsymbol{X}\|_2^2 + \alpha\|Q-TX\|_2^2 \quad \text{s.t. } \|\boldsymbol{x}\|_0 \le S. \tag{2}$$

The newly introduced term, i.e., $\|\boldsymbol{Q}-\boldsymbol{T}\boldsymbol{X}\|_2^2$, represents discriminative sparse-code error, where $\boldsymbol{Q}=[\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_N] \in R^{M\times N}$ is used to control sparse mode. $\boldsymbol{q}_i=[q_i^1, q_i^2, \ldots, q_i^M]^T=[0\ldots1,1,\ldots0]^T \in R^M$ is related to feature sample $\boldsymbol{f}_i$: when $\boldsymbol{f}_i$ is of the same class with the mth dictionary atom a_m, $q_i^M=1$; otherwise, $q_i^M=0$. For example, assuming $\boldsymbol{A}=[\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3, \boldsymbol{a}_4, \boldsymbol{a}_5, \boldsymbol{a}_6]$ and $\boldsymbol{F}=[\boldsymbol{f}_1, \boldsymbol{f}_2, \boldsymbol{f}_3, \boldsymbol{f}_4]$, where

f_1, a_1, a_2 are from class 1; f_2, a_3, a_4 are from class 2; and f_3, f_4, a_5, a_6 are from class 3, Q can be defined as

$$Q \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

$T \in R^{M\times M}$ is a matrix representing a linear transform that converts original sparse coefficients matrix $\boldsymbol{X}$ to be most discriminative.

This model can be expressed in an equivalent form, i.e.,

$$\langle \boldsymbol{A}, \boldsymbol{T}, \boldsymbol{X} \rangle = \arg \min_{\boldsymbol{A},\boldsymbol{F},\boldsymbol{X}} \left\| \begin{pmatrix} \boldsymbol{F} \\ \sqrt{\alpha}\boldsymbol{Q} \end{pmatrix} - \begin{pmatrix} \boldsymbol{A} \\ \sqrt{\alpha}\boldsymbol{T} \end{pmatrix} \boldsymbol{X} \right\|_2^2$$
$$\text{s.t. } \forall\, i, \|\boldsymbol{x}_i\|_0 \le S. \tag{3}$$

Let

$$\boldsymbol{F}_{\text{new}} = (\boldsymbol{F}^T, \sqrt{\alpha}\boldsymbol{Q}^T)^T$$
$$\boldsymbol{A}_{\text{new}} = \left(\boldsymbol{A}^T, \sqrt{\alpha}\boldsymbol{T}^T\right)^T. \tag{4}$$

Then, (3) can be reformulated as

$$\langle \boldsymbol{A}_{\text{new}}, \boldsymbol{X} \rangle = \arg \min_{A_{\text{new}}, X} \|\boldsymbol{F}_{\text{new}} - \boldsymbol{A}_{\text{new}}\boldsymbol{X}\|_2^2$$
$$\text{s.t. } \forall\, i, \|\boldsymbol{x}_i\|_0 \le S. \tag{5}$$

This way, the model can be solved by exploiting the traditional K-SVD algorithm.

Here, $\boldsymbol{Q}$, which represents the label of the dictionary atom, is a constant matrix. It constrains that the labels of the learned dictionary atoms are the same with those of the initial dictionary atoms.

In this letter, LC-KSVD is used to learn dictionaries for both features, and the learned dictionaries will be used in the following JDSR method.

C. Joint Dynamic Sparse Representation based ATR

Given two kinds of features extracted from a single image, we can put the two sparse representation problems together as

$$\hat{\boldsymbol{X}} = \arg \min_X \sum_{k=1}^{2} \|\boldsymbol{f}^k - \boldsymbol{A}^k \boldsymbol{x}^k\|_2^2$$
$$\text{s.t. } \|\boldsymbol{x}^k\|_0 \le S, \forall\, 1 \le k \le 2 \tag{6}$$

where $\hat{\boldsymbol{X}} = [\boldsymbol{x}^1, \boldsymbol{x}^2]$ is the sparse coefficient matrix for a test sample $\boldsymbol{F} = [\boldsymbol{f}^1, \boldsymbol{f}^2]$, and A^k is the dictionary for the kth feature, which is learned via (3).

In (6), however, the sparse coefficients of two features are independently computed. To combine different features, the sparse coefficients of different features are simultaneously computed by the JDSR model [9]. A new term, i.e., dynamic active set, $\boldsymbol{g}_s \in R^2$, $s = 1, 2, \ldots, S$, is introduced to express the correlations of different features. As shown in Fig. 2, in the sparse coefficient matrix $\boldsymbol{X}$, each column denotes a sparse representation vector for the corresponding feature. Squared blocks denote coefficient values, where a white block denotes a zero entry value. Each dynamic active set refers to the row indices of a set of coefficients belonging to the same class, e.g., $\boldsymbol{g}_1 = [2, 3]$. During the JSR process, a number of dynamic active sets are jointly activated. l_2-*norm* is applied over each dynamic set $\boldsymbol{g}_s$ to combine the strength of all the atoms within a dynamic set and l_0- *norm* across l_2-*norm* is applied to ensure that only a small number of dynamic sets be involved in JSR. Therefore, the promoting term of the JDSR model can be expressed as follows:

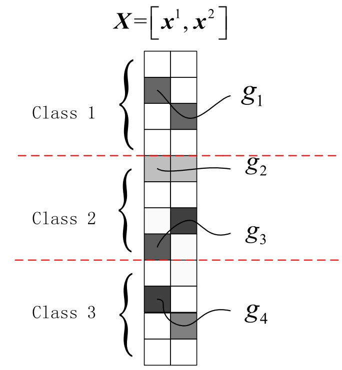

Fig. 2. Illustration of the dynamic active sets.

$$\|\boldsymbol{X}\|_G = \|[\|x_{g_1}\|_2, \|x_{g_2}\|_2 \cdots]\|_0 \tag{7}$$

where $\boldsymbol{x}_{g_s} = \boldsymbol{X}(\boldsymbol{g}_s) = [\boldsymbol{X}(\boldsymbol{g}_s(1), 1), \boldsymbol{X}(\boldsymbol{g}_s(2), 2)]^T \in R^2$ refers to the vector formed as the collection of the coefficients related to the sth dynamic active set $\boldsymbol{g}_s$.

Then, the JDSR model can be reformulated as

$$\hat{\boldsymbol{X}} = \arg \min_X \sum_{k=1}^{2} \|\boldsymbol{f}^k - \boldsymbol{A}^k \boldsymbol{x}^k\|_2^2$$
$$\text{s.t. } \|X\|_G \le S. \tag{8}$$

For a test sample $\boldsymbol{F} = [\boldsymbol{f}^1, \boldsymbol{f}^2]$, we use minimum reconstruction residual as the criterion to estimate the class label, which can be formulated as

$$\hat{c} = \arg \min_c \sum_{k=1}^{2} \omega^k \left\|\boldsymbol{f}^k - \boldsymbol{A}^k \delta_c(\boldsymbol{x}^k)\right\|_2^2 \tag{9}$$

where $\hat{c}$ is the label of the test sample, and ω^k is the reconstruction parameter for the kth feature. The operator $\delta_c(\cdot)$ denotes the operation of preserving the rows that corresponds to class c and setting all others to be zero.

In this letter, the amplitude feature and the SIFT feature of a test image data are jointly represented with the JDSR model and the test image is recognized according to reconstruction residual.

III. Experimental Results

Here, we demonstrate the performance of our proposed method for SAR ATR by presenting numerical recognition results on the MSTAR public data set.

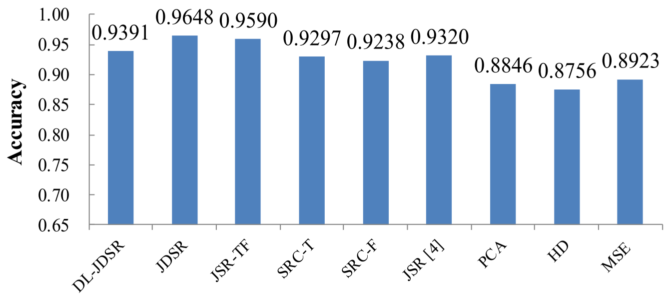

Fig. 3. Recognition accuracies of different methods for the 3-class data.

A. Introduction of MSTAR Data Set

We validate our method on MSTAR data set. The data set consists of many SAR targets under different aspect angles and pitch angles. The resolution for each image is 0.3 m $\times$ 0.3 m. We carry out the experiments on both 3-class data and 10-class data, among which the target data at depression angle 17° are taken as the training data, whereas the target data at depression angle 15° are taken as the test data. The aspect angles of every kind of targets range from 0° to 360°.

The 3-class data contains the targets of BMP2, BTR70, and T72; and they have 7 subtypes in total. The details about the selection of training and test samples are the same as [1].

Aside from the 3-class data, the 10-class data contains the data from another 7 ground vehicles. The details about selection of training and test samples are the same as [14].

B. Recognition Results

In the following experiments, aside from our DL-JDSR, we also implement three SAR ATR methods for comparison, which are the degraded versions of DL-JDSR. The first two are the ATR methods based on JDSR and JSR with both of the image domain amplitude feature and the SIFT feature (referred to as "JDSR" and "JSR-TF" in the following experiments), which use both of the feature matrixes of the training data, i.e., $\boldsymbol{F}^1$ and $\boldsymbol{F}^2$, as the dictionaries in the training stage. The other two are the ATR methods based on sparse representations, respectively, with the image domain amplitude feature and the SIFT feature (referred to as "SRC-T" for the image domain amplitude feature and "SRC-F" for the SIFT feature in the following experiments, respectively), which use the image domain amplitude feature matrix of the training data $\boldsymbol{F}^1$ and the SIFT feature matrix of the training data $\boldsymbol{F}^2$ as the dictionaries in the training stages, respectively, and use the sparse representation algorithms [3] in the test stages. In all these methods, we set reconstruction parameters $\omega = [0.7, 0.3]$.

1) Three-Class Data Recognition Results: In Fig. 3, we compare the recognition results of our DL-JDSR, JDSR, JSR-TF, SRC-T, and SRC-F with those of some existing SAR recognition methods for the 3-class data, which includes JSR with multiview SAR images [4] (referred to as "JSR" in Fig. 3), PCA with kernel SVM [1] (referred to as "PCA" in Fig. 3), particle swarm optimization with Hausdorff distance [15] (referred to as "HD" in Fig. 3), and template matching [2] (referred to as "MSE" in Fig. 3). In this letter, we set the sparse level $S = 15$ for DL-JDSR, JDSR, SRC-T, and SRC-F. As shown in Fig. 3, JDSR, JSR-TF, and DL-JDSR yield the highest three accuracies of 0.9648, 0.9590, and 0.9391, due to the combination of the amplitude feature and the SIFT feature. The result of JDSR is better than that of JSR-TF, which proves that the JDSR not only enables a more flexible atom selection process but also enhances the recognition accuracy. Additionally, although the recognition accuracy of DL-JDSR is lower than those of JDSR and JSR-TF, the dictionaries of DL-JDSR $\boldsymbol{A}^1 \in R^{d\times 138}$ and $\boldsymbol{A}^2 \in R^{d\times 138}$ are much smaller than those of JDSR and SRC-TF $\boldsymbol{F}^1 \in R^{d\times 698}$ and $\boldsymbol{F}^2 \in R^{d\times 698}$ in the experiment in Fig. 3. The comparison for the three methods with the same dictionary size will be given later.

TABLE I
CONFUSION MATRIX OF DL-JDSR FOR THE 3-CLASS DATA

	BMP2	BTR70	T72
BMP9563	**0.9436**	0.0513	0.0051
BMP9566	**0.9031**	0.0612	0.0357
BMPC21	**0.9439**	0.0459	0.0102
BTRC71	0.0102	**0.9898**	0
T72132	0.0153	0.0051	**0.9796**
T72S7	0.0314	0	**0.9686**
T72812	0.1282	0.0256	**0.8462**

The confusion matrix of DL-JDSR for the 3-class data is given in Table I.

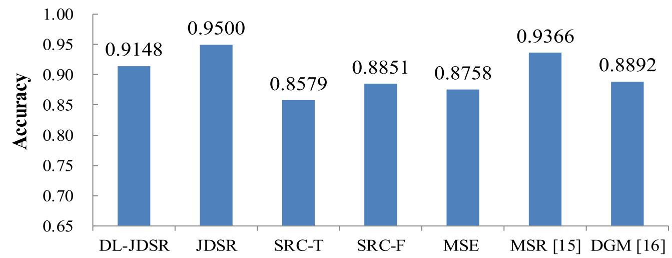

Fig. 4. Recognition accuracies of different methods for the 10-class data.

2) Ten-Class Data Recognition Results: In Fig. 4, we compare the recognition results of our DL-JDSR, JDSR, SRC-T, and SRC-F with those of some existing SAR recognition methods for the 10-class data, which includes template matching [2] (referred to as "MSE" in Fig. 4), sparse representation of monogenic signal [14] (referred to as "MSR" in Fig. 4), and discriminative graphic model [16] (referred to as "DGM" in Fig. 4). In this letter, we set the sparse level $S = 15$ for DL-JDSR, JDSR, SRC-T, and SRC-F. As shown in Fig. 4, for the case that all the training samples are used as dictionary atoms, i.e., $F^1 \in R^{d\times 2746}$ and $F^2 \in R^{d\times 2746}$, JDSR yields the highest accuracy of 0.9500. Although the recognition accuracy of DL-JDSR is lower than that of JDSR, the dictionaries of DL-JDSR, $A^1 \in R^{d\times 681}$ and $A^2 \in R^{d\times 681}$, are much smaller. Moreover, when the dictionary size of JDSR is reduced to 681, the recognition accuracy decreases to 0.8745, which is lower than that of DL-JDSR (0.9148). By the way, MSR achieves an impressive performance (0.9366) due to its usage of the complex data, which contains both the amplitude and the phase information. The confusion matrix of DL-JDSR for the 10 class data is given in Table II.

3) Effectiveness of Dictionary Learning: Furthermore, to examine the effectiveness of dictionary learning, we compare

TABLE II
CONFUSION MATRIX OF DL-JDSR FOR THE 10-CLASS DATA

	BMP2	BTR70	T72	2S1	BRDM	BTR60	D7	T62	ZIL131	ZSU234
BMP9563	**0.9231**	0.0103	0.0256	0.0103	0.0051	0	0.0051	0.0051	0.0051	0.0103
BMP9566	**0.8265**	0.0153	0.0510	0.0459	0.0153	0.0102	0	0.0051	0.0051	0.0255
BMPC21	**0.9133**	0.0051	0.0153	0.0153	0.0102	0	0.0051	0.0153	0.0051	0.0153
BTRC71	0	**0.9388**	0	0	0.0459	0.0051	0	0.0051	0.0051	0
T72132	0	0	**0.9541**	0	0	0.0153	0.0102	0.0153	0	0.0051
T72S7	0.0105	0.0052	**0.8953**	0.0209	0.0052	0.0052	0	0.0157	0.0105	0.0314
T72812	0.0615	0.0051	**0.7385**	0.0513	0	0.0103	0.0256	0.0667	0.0205	0.0205
2S1	0.0365	0.0036	0	**0.8905**	0.0109	0.0109	0.0036	0.0109	0.0146	0.0182
BRDM	0	0.0182	0	0.0146	**0.9161**	0.0219	0.0036	0.0036	0.0146	0.0073
BTR60	0.0051	0.0103	0.0154	0	0.0513	**0.8821**	0.0051	0.0051	0	0.0256
D7	0	0	0	0.0036	0	0	**0.9854**	0.0073	0	0.0036
T62	0	0	0	0.0110	0.0037	0	0.0037	**0.9670**	0.0037	0.0110
ZIL131	0	0	0	0.0292	0.0036	0.0036	0.0073	0.0292	**0.9234**	0.0036
ZSU234	0	0	0	0	0.0073	0.0036	0	0	0.0073	**0.9818**

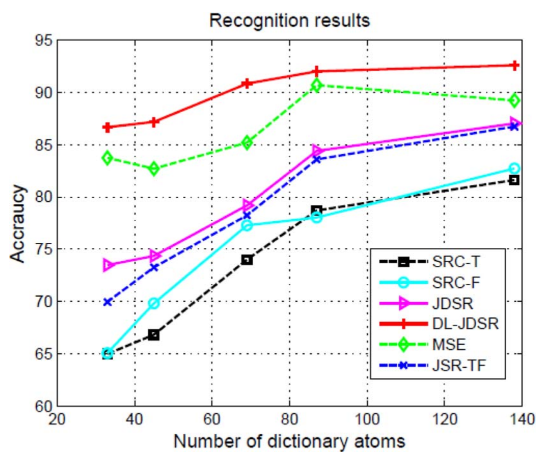

Fig. 5. Variation of recognition performance with number of dictionary atoms.

the variations of the recognition performance with the number of dictionary atoms via DL-JDSR, JDSR, JSR-TF, SRC-T and SRC-F and template matching (referred to as "MSE" in Fig. 5) for the 3-class data. In this letter, the sparse level $S = 5$. To adjust the number of dictionary atoms, one from every n training samples is selected for training in the experiment with $n = 5,8,10,15,20$. For JDSR, SRC-TF, SRC-T, and SRC-F, the feature vectors of the selected training samples are directly used as dictionary atoms, whereas for DL-JDSR, the dictionary is learned via the LC-KSVD algorithm, which uses the training feature vectors as the initial dictionary atoms in its learning process.

As shown in Fig. 5, the recognition accuracies of DL-JDSR are much higher than those of JDSR, SRC-TF, SRC-T, SRC-F and MSE, particularly in the cases with a small number of dictionary atoms.

IV. CONCLUSION

In this letter, a novel SAR image recognition method based on dictionary learning and JDSR is proposed. The method exploits the LC-KSVD algorithm to learn compact and distinctive dictionaries. In addition, it not only takes advantage of correlations between two features, i.e., the image domain amplitude feature and the SIFT feature, of a single SAR image but also enables a more flexible and adaptive atom selection process for JSR. Experiment results show that the proposed method has a better performance than other existing methods on MSTAR data set.

REFERENCES

[1] Y. Wang, P. Han, X. Lu, R. Wu, and J. Huang, "The performance comparison of Adaboost and SVM Applied to SAR ATR," in *Proc. IEEE Int. Radar Conf.*, 2006, pp. 1–4.
[2] K. Ikeuchi, T. Shakunaga, M. D. Wheeler, and T. Yamazaki, "Invariant histograms and deformable template matching for SAR target recognition," in *Proc. CVPR*, 1996, pp. 100–105.
[3] J. Wright, A. Y. Yang, A. Ganesh, S. Shankar Sastry, and Y. Ma, "Robust face recognition via sparse representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 2, pp. 210–227, Feb. 2009.
[4] H. Zhang, M. Nasser, Y. Zhang, and T. S. Huang, "Multi-view automatic target recognition using joint sparse representation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 3, pp. 2481–2497, Jul. 2012.
[5] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *IEEE Int. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.
[6] Q. Yu, B. Pang, P. Wu, and Y. Zhang, "Automatic coarse-to-precise subpixel multi-band SAR images co-registration based affine SIFT and Radial Base Function (RBF)," in *Proc. IEEE 5th APSAR*, 2015, pp. 684–687.
[7] M. Aharon, M. Elad, and A. Bruckstein, "K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Trans. Signal Process.*, vol. 54, no. 1, pp. 4311–4322, Nov. 2006.
[8] Z. L. Jiang, Z. Lin, and S. Davis Larry, "Label consistent K-SVD: Learning a discriminative dictionary for recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2651–2664, Nov. 2013.
[9] H. Zhang, M. Nasser, Y. Zhang, and T. S. Huang, "Multi-observation visual recognition via joint dynamic sparse representation," in *Proc. IEEE Int. Comput. Vis.*, 2011, pp. 595–602.
[10] R. Weisenseel, W. Carl, D. Castanon, and R. Brewer, "MRF-based algorithms for segmentation of SAR images," in *Proc. IEEE Int. Comput. Vis. Image Process.*, 1998, pp. 770–774.
[11] Y. Chao and D. Casasentstal, "MSTAR 10-class classification and confuser and clutter rejection using SVRDM," in *Proc. SPIE Opt. Pattern Recognit. XVII*, 2006, vol. 6245, pp. 1–13.
[12] S. Lazebnik, C. Schmid, and J. Ponce, "Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories," in *Proc. CVPR*, 2006, pp. 2169–2178.
[13] L. Fei-Fei and P. Perona, "A Bayesian hierarchical model for learning natural scene categories," in *Proc. CVPR*, 2005, pp. 1–8.
[14] G. Dong, G. Kuang, and N. Wang, "SAR target recognition via joint sparse representation of monogenic signal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, pp. 3316–3328, no. 7, Jul. 2015.
[15] Dungan, K. Edward and R. L. Moses, "Feature-based vehicle classification in wide-angle synthetic aperture radar," Ph.D. dissertation, Electr. Comput. Eng., Ohio State Univ., Columbus, OH, USA, 2010.
[16] U. Srinival, V. Monga, and R. G. Raj, "SAR automatic target recognition using discriminative graphical models," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 1, pp. 591–606, Jan. 2014.